# Solution overview: Credit Risk Portfolio Monitor

Streamlit dashboard over 5 synthetic bank tables (14,685 loan accounts, 10,000 customers). Repo layout and run commands are in README.md.

## What the dashboard shows and how an analyst uses it

| | |
|---|---|
| **Portfolio** | KPIs; exposure and actual default rate by model risk band (A-E); exposure by loan type x band; default rate, PD and exposure by any dimension; cross-cut heatmap (any two dimensions); vintage and month-on-book trends; drill-down from a segment to its accounts. Sidebar filters (loan type, region, employment, income, band, year) apply throughout. Use: find where risk and money concentrate, then drill in. |
| **Account** | PD, band, percentile, top risk drivers for the account, profile, 12-month EMI/DPD chart, bureau lines, collections, anomaly reasons. Use: explain a score before a limit, collections or override decision. |
| **Ask the Portfolio** | Plain-English questions with exact figures, the scope used, the result table and the query spec. |
| **Anomaly review** | Prioritised queue of flagged accounts, why each was flagged, peer comparison, decision + log. |
| **Model / Monitoring** | Metrics, ROC, deciles, importance; monitoring health check (PSI, cohort AUC/KS, calibration by band, RAG thresholds) and a drift simulator. |
| **Data & method** | Join logic and the full data-quality log. |

## The two AI features

| | Ask the Portfolio (LLM) | Anomaly detection |
|---|---|---|
| **How it works** | LLM call 1 turns the question into a JSON query using only the schema (no rows). Code validates it against a whitelist and computes the answer with pandas. LLM call 2 words a 2-3 sentence answer from the aggregated result; every number is checked against the result, otherwise a template answer is used. Out-of-scope questions (forecasts, 'why', outside facts, personal data) are declined. Works without a key via a keyword parser. | Three detectors, no default labels: (1) own history: months 10-12 vs the account's months 1-9 (new missed EMIs, DPD jump of 4+ of its own SDs, 50-pt fall in share paid); (2) peer segment: robust z >= 4.5 vs loan type x income band on EMI/income, bureau utilisation, debt/income, new lines, write-offs; (3) Isolation Forest, top 1.5%. 654 accounts flagged; back-test default rate 28.4% vs 15.5% unflagged. |
| **Trigger** | Analyst submits a question. | Every build (nightly in production) over all accounts. |
| **False positive** | A confident answer on the wrong slice (e.g. a synonym mapped to the wrong column). | A harmless outlier, e.g. high EMI/income with income not captured, or a one-off missed EMI. |
| **False negative** | A valid question declined or mis-parsed. | Slow deterioration that stays inside own and peer norms; new borrowing after the last bureau pull. |
| **Human review** | Scope line, result table and query spec shown under each answer; thumbs feedback. | Analyst records Confirm / Dismiss / Need info with reason; no customer action without it; dismiss reasons retune thresholds. |

## Model results and approach

| Model | Holdout AUC | KS | Gini | 5-fold CV AUC | Out-of-time AUC (2023) |
|---|---|---|---|---|---|
| Logistic regression (champion) | 0.682 | 0.268 | 0.364 | 0.681 ± 0.005 | 0.695 |
| LightGBM | 0.676 | 0.268 | 0.351 | 0.679 ± 0.009 | 0.687 |

35 numeric features (12-month repayment behaviour, bureau aggregates, affordability ratios, age, income, missing flags) + loan type and employment type. Seven weak categoricals (marital status, education, housing, purpose, channel, region, city tier) were dropped: CV AUC rose 0.676 -> 0.681 and fair-lending risk fell; gender never used. Collections excluded (leakage). Split grouped by customer; champion by grouped-CV AUC: the logistic regression, an explainable scorecard; LightGBM kept as challenger. Top drivers: income, late payments, loan amount, bureau lines and DPD, years employed, age. Leakage check (bureau lines opened before origination only): AUC 0.671. Monitoring page: PSI, cohort AUC/KS, calibration by band (band A currently amber, under-predicts), fairness by segment, drift simulator. Expected loss = PD x assumed LGD x exposure.

## Top 3 portfolio insights

**1. Mortgages carry most of the risk in money terms.** Mortgages are 10% of accounts but 58% of exposure and 70% of defaulted exposure. Their default rate is 24.7% against 15.2% for all other products. *So what:* Exposure-weighted, the portfolio's risk is a mortgage question. Prioritise a mortgage underwriting review and request LTV and property values, which are missing from this data, so loss given default can be estimated.

**2. Income is the clearest single driver.** Default rate falls steadily with income, from 25.7% below $58k to 9.5% above $101k (2.7x). The lowest band holds 20% of exposure but 30% of defaulted exposure. *So what:* Affordability should drive limits: cap EMI-to-income for the lowest band and route sub-$58k applications for larger loans to manual review.

**3. Unemployed, low-income borrowers are the worst segment (cross-cut).** Unemployed borrowers earning under $58k default at 38.4% (531 accounts), against 9.1% for salaried borrowers above $101k. Unemployed borrowers are 5% of exposure but 11% of defaulted exposure. Low-income mortgages default at 35.4%. *So what:* A combined policy rule (employment status x income) is sharper than either cut-off alone. Require a guarantor or income proof for unemployed applicants and review the existing low-income mortgage book first.

## Key assumptions

Data is a snapshot at 2025-12-31; exposure = outstanding balance after month 12; target_default is final and its observation window is unknown; bureau is the latest snapshot (a behavioural, monitoring-style score), checked against a point-in-time variant; 7,293 loans predating customer_since, 3,875 collections events before origination and 3,149 future close dates are treated as data errors and flagged, not dropped.

## What I would need from the bank's IT and data teams for production

(1) Governed feeds from core banking, the bureau and the collections system into the warehouse, with point-in-time bureau archives and a defined default definition and outcome window. (2) A scheduled pipeline (Airflow/dbt) with data-quality tests that block bad loads. (3) Model registry, validation sign-off and a scoring service. (4) An approved LLM endpoint inside the bank's cloud tenancy with no data retention, plus logging of prompts and answers. (5) SSO and role-based access for the dashboard, and a decisions table in the database instead of a CSV. (6) Model risk policy for monitoring thresholds, fairness review and retraining.